

WBJEE Mock Test: Chemistry

Total Marks: 100 | Time Allowed: 2 Hours

General Instructions

1. This paper contains 75 questions divided into three categories
2. **Category-I (Q1 - Q50):** 1 Mark each. Only one option is correct. Negative marking is -0.25 marks.
3. **Category-II (Q51 - Q65):** 2 Marks each. Only one option is correct. Negative marking is -0.5 marks.
4. **Category-III (Q66 - Q75):** 2 Marks each. One or more options may be correct. No negative marking. Partial marks are awarded for partially correct answers.

CATEGORY - I (1 Mark Each)

Only one option is correct. Correct response yields 1 mark. Incorrect response yields -0.25 marks.

1. In SO_2 , NO_2^- , and N_3^- , the hybridizations at the central atom are respectively
A) sp^2 , sp^2 and sp
B) sp^2 , sp and sp
C) sp^2 , sp^2 and sp^2
D) sp , sp^2 and sp
Ans: A
2. The type of hybridization and the magnetic property of $[\text{MnCl}_6]^{3-}$ are
A) d^2sp^3 , paramagnetic (4 unpaired e^-)
B) sp^3d^2 , paramagnetic (4 unpaired e^-)
C) d^2sp^3 , paramagnetic (2 unpaired e^-)
D) sp^3d^2 , paramagnetic (2 unpaired e^-)
Ans: B
3. Which of the following species is diamagnetic according to Molecular Orbital Theory?
A) O_2
B) B_2
C) C_2
D) N_2^+
Ans: C
4. The SI unit of molar conductivity is
A) $\text{S cm}^2 \text{ mol}^{-1}$
B) $\text{S m}^2 \text{ mol}^{-1}$
C) $\Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$
D) $\Omega \text{ m mol}^{-1}$
Ans: B

5. The entropy of a perfect crystal at 0 K is

- A) Maximum
- B) Minimum but not zero
- C) Zero
- D) Infinite

Ans: C

6. The pH of pure water at 25°C is

- A) 0
- B) 14
- C) 7
- D) 1

Ans: C

7. Rate of diffusion of gases is governed by

- A) Boyle's law
- B) Charles' law
- C) Graham's law
- D) Dalton's law

Ans: C

8. The Bond Order of He_2^+ is:

- A) 0
- B) 0.5
- C) 1
- D) 1.5

Ans: B

9. Which of the following molecules has a square pyramidal geometry?

- A. PCl_5
- B. BrF_5
- C. XeF_2
- D. SF_6

Ans: B

10. The increasing order of dipole moment for the following is:

- A. $\text{CH}_4 < \text{NH}_3 < \text{H}_2\text{O}$
- B. $\text{H}_2\text{O} < \text{NH}_3 < \text{CH}_4$
- C. $\text{NH}_3 < \text{CH}_4 < \text{H}_2\text{O}$
- D. $\text{CH}_4 < \text{H}_2\text{O} < \text{NH}_3$

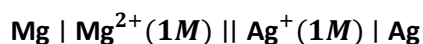
Ans: A

11. A reaction whose rate is independent of concentration is of

- A) First order
- B) Second order
- C) Zero order
- D) Third order

Ans: C

12. For the cell:



Standard reduction potentials are:

- $\text{Mg}^{2+} + 2\text{e}^{-} \rightarrow \text{Mg} \quad E^{\circ} = -2.37 \text{ V}$
- $\text{Ag}^{+} + \text{e}^{-} \rightarrow \text{Ag} \quad E^{\circ} = +0.80 \text{ V}$

The standard EMF of the cell is:

- A) 1.57 V
- B) 2.37 V
- C) 3.17 V
- D) -3.17 V

Ans: C

13. Which of the following is a state function?

- A) Heat
- B) Work
- C) Internal energy
- D) Path length

Ans: C

14. 22.4 L of an ideal gas at STP contains

- A) 6.022×10^{23} molecules
- B) 1 gram
- C) 0.5 mole
- D) 2 moles

Ans: A

15. One mole of a real gas obeys the Van der Waals equation at a certain temperature, the values are:

- i. $a = 3.6 \text{ L}^2 \text{ atm mol}^{-2}$
- ii. $b = 0.04 \text{ L mol}^{-1}$
- iii. $T = 300 \text{ K}$
- iv. $V = 10 \text{ L}$
- v. $R = 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$

The pressure of the gas is

- A) 2.36 atm
- B) 2.46 atm
- C) 2.56 atm
- D) 2.66 atm

Ans: B

16. Standard enthalpy of formation of an element in its standard state is

- A) 1
- B) Zero
- C) -1
- D) 100 kJ

Ans: B

17. A process is spontaneous if

- A) $\Delta H > 0$
- B) $\Delta S < 0$
- C) $\Delta G < 0$
- D) $\Delta G > 0$

Ans: C

18. Hybridization of carbon in CO_2 is

- A) sp^3
- B) sp^2
- C) sp
- D) dsp^2

Ans: C

19. SI unit of surface tension is

- A) N m^{-2}
- B) N m^{-1}
- C) J m^{-1}
- D) Pa

Ans: B

20. The Arrhenius equation relates rate constant with

- A) Pressure
- B) Volume
- C) Temperature
- D) Concentration

Ans: C

21. Which of the following molecules involves sp^3d hybridization and has different bond lengths?

- A. XeF_4
- B. BF_3
- C. SF_4
- D. CH_4

Ans: C

22. According to VSEPR theory, the geometry of XeOF_4 is:

- A. Octahedral
- B. Square Pyramidal
- C. Trigonal Bipyramidal
- D. Square Planar

Ans: B

23. Which gas shows maximum deviation from ideal behaviour at high pressure?

- A) H_2
- B) He
- C) SO_2
- D) N_2

Ans: C

24. The atomic number of the element with configuration $[\text{Ne}] 3s^2 3p^3$ is

- A) 14
- B) 15
- C) 16
- D) 13

Ans: B

25. Which of the following is diamagnetic?

- A) O_2
- B) B_2
- C) C_2
- D) NO

Ans: C

26. Which pair of ions has the same number of unpaired electrons?

- A. Ti^{3+} and V^{3+}
- B. V^{3+} and Ni^{2+}
- C. Fe^{3+} and Mn^{2+}
- D. Both B and C

Ans: D

27. The hybridization of Nickel in $[\text{Ni}(\text{CN})_4]^{2-}$ is:

- A. sp^3
- B. dsp^2
- C. sp^3d^2
- D. d^2sp^3

Ans: B

28. The most electronegative element is

- A) Oxygen
- B) Nitrogen
- C) Fluorine
- D) Chlorine

Ans: C

29. Oxidation state of Cr in $K_2Cr_2O_7$ is

- A) +3
- B) +6
- C) +2
- D) +7

Ans: B

30. Which species has maximum number of unpaired electrons?

- A. O_2
- B. O_2^+
- C. O_2^-
- D. N_2

Ans: A

31. Shape of NH_3 molecule is

- A) Trigonal planar
- B) Tetrahedral
- C) Trigonal pyramidal
- D) Linear

Ans: C

32. Noble gas with lowest ionization energy is

- A) He
- B) Ne
- C) Ar
- D) Rn

Ans: D

33. Chief ore of aluminium is

- A) Hematite
- B) Galena
- C) Bauxite
- D) Magnetite

Ans: C

34. Which complex is most likely to be high spin?

- A. $[Fe(CN)_6]^{4-}$
- B. $[Fe(H_2O)_6]^{2+}$
- C. $[Co(NH_3)_6]^{3+}$

D. $[\text{Cr}(\text{CN})_6]^{3-}$

Ans: B

35. Central metal ion present in hemoglobin is

A) Mg^{2+}

B) Fe^{2+}

C) Cu^{2+}

D) Zn^{2+}

Ans: B

36. Electrolyte used in lead storage battery is

A) HCl

B) HNO_3

C) H_2SO_4

D) H_3PO_4

Ans: C

37. Highest first ionization energy in second period is of:

A) F

B) N

C) O

D) Ne

Ans: D

38. Coordination number of sulfur in SF_6 is:

A) 4

B) 5

C) 6

D) 2

Ans: C

39. The correct order of bond length is:

A) $\text{O}_2^+ < \text{O}_2 < \text{O}_2^-$

B) $\text{O}_2 < \text{O}_2^+ < \text{O}_2^-$

C) $\text{O}_2^- < \text{O}_2 < \text{O}_2^+$

D) $\text{O}_2 < \text{O}_2^- < \text{O}_2^+$

Ans: A

40. The correct order of electron affinity is

A. $\text{F} > \text{Cl} > \text{Br} > \text{I}$

B. $\text{Cl} > \text{F} > \text{Br} > \text{I}$

C. $\text{Cl} > \text{Br} > \text{F} > \text{I}$

D. $\text{F} > \text{Cl} > \text{I} > \text{Br}$

Ans: B

41. The standard electrode potential of a half-cell depends on:

A) Nature of electrode

B) Concentration of ions

- C) Temperature
- D) All of these

Ans: D

42. Hybridization of carbon in ethene is

- A) sp
- B) sp^2
- C) sp^3
- D) dsp^2

Ans: B

43. Alkene reacts with cold alkaline $KMnO_4$ to form

- A) Alcohol
- B) Ketone
- C) Vicinal diol
- D) Alkane

Ans: C

44. Which hydrocarbon burns with a sooty flame?

- A) Alkane
- B) Alkene
- C) Alkyne
- D) Aromatic hydrocarbon

Ans: D

45. Monomer of PVC is

- A) Ethene
- B) Vinyl chloride
- C) Styrene
- D) Tetrafluoroethylene

Ans: B

46. Reaction of alkyl halide with Na in dry ether gives alkane. This is called

- A) Fittig reaction
- B) Wurtz reaction
- C) Kolbe reaction
- D) Sandmeyer reaction

Ans: B

47. Which functional group undergoes nucleophilic addition?

- A) Alkene
- B) Alkane
- C) Carbonyl
- D) Ether

Ans: C

48. Which pair represents enantiomers?

- A) (R)-2-butanol & (S)-2-butanol
- B) cis-2-butene & trans-2-butene

- C) 1-butanol & 2-butanol
D) meso-tartaric acid & d-tartaric acid

Ans: A

49. Glucose is a

- A) Protein
B) Lipid
C) Carbohydrate
D) Vitamin

Ans: C

50. Polymer used in non-stick cookware is

- A) PVC
B) Nylon
C) Bakelite
D) Teflon

Ans: D

CATEGORY - II (2 Marks Each)

Only one option is correct. Correct response yields 2 marks. Incorrect response yields -0.5 marks.

51. Which isomerism is shown by Diethyl ether and Methyl propyl ether?

- A. Chain isomerism
B. Metamerism
C. Functional isomerism
D. Geometrical isomerism

Answer: B

52. Among the following, the extensive variables are

- (A) V (Volume)
(B) P (Pressure)
(C) E (Internal energy)
(D) H (Enthalpy)

Ans: A,C,D

53. X is an extensive property, and x is an intensive property of a thermodynamic system. Which of the following statement(s) is (are) correct?

- (A) xX is extensive
(B) x/X is intensive
(C) X/x is extensive
(D) dX/dx is intensive

Ans: A,C

54. For spontaneous polymerization, which of the following is (are) correct?

- (A) ΔG is negative
- (B) ΔH is negative
- (C) ΔS is positive
- (D) ΔS is negative

Ans: A,B,D

55. Which of the following compounds will show a positive result with Tollen's reagent?

- A. Glucose
- B. Fructose
- C. Benzaldehyde
- D. Acetone

Answers: A, B, C

56. Which of the following will undergo a Cannizzaro reaction when treated with concentrated NaOH?

- A. Formaldehyde
- B. Benzaldehyde
- C. Acetaldehyde
- D. Trimethyl acetaldehyde

Answers: A, B, D

57. Which of the following statements about S_N1 reactions are correct?

- A. It follows first-order kinetics.
- B. It involves the formation of a carbocation intermediate.
- C. It is favoured by polar protic solvents.
- D. It results in complete inversion of configuration.

Answers: A, B, C

58. Which of the following are examples of addition polymers?

- A. Polyethylene
- B. PVC
- C. Nylon-6,6
- D. Teflon

Answers: A, B, D

59. Which of the following compounds will give a yellow precipitate with Iodine and NaOH (Iodoform test)?

- A. Ethanol
- B. Acetophenone
- C. Benzophenone
- D. Isopropyl alcohol

Answers: A, B, D

60. Which of the following reagents can be used to distinguish between Ethene and Ethyne?

- A. Bromine water
- B. Ammoniacal Cuprous Chloride
- C. Ammoniacal Silver Nitrate
- D. Alkaline $KMnO_4$

Answers: B, C

61. Which of the following are periodic properties that generally decrease down a group in the periodic table?

- A. Atomic radius
- B. Electronegativity
- C. Ionization enthalpy
- D. Electron gain enthalpy (less negative)

Answers: B, C, D

62. Which of the following undergo nucleophilic substitution (S_N^1) faster than Ethyl Chloride?

- A. Benzyl Chloride
- B. Tert-butyl Chloride
- C. Allyl Chloride
- D. Chlorobenzene

Answers: A, B, C

63. Which of the following are examples of intensive properties?

- A. Temperature
- B. Density
- C. Molar Heat Capacity
- D. Volume

Answers: A, B, C

64. Which of the following properties are characteristic of transition elements?

- A. Formation of coloured ions
- B. Variable oxidation states
- C. Paramagnetic behaviour
- D. Formation of interstitial compounds

Answers: A, B, C, D

65. Which of the following compounds are aromatic in nature?

- A. Benzene
- B. Naphthalene
- C. Anthracene
- D. Cyclooctatetraene

Answers: A, B, C

CATEGORY - III (2 Marks Each)

*One or more options may be correct. Correct response yields 2 marks.
No negative marking.*

66. In which of the following reactions is the major product formed according to Saytzeff's rule?

- A. Dehydration of 2-Butanol
- B. Dehydrohalogenation of 2-Bromobutane with Alc. KOH
- C. Dehydrohalogenation of 2-Bromobutane with t-BuOK
- D. Pyrolysis of quaternary ammonium hydroxides

Answers: A, B

67. Which of the following statements about Enzymes are correct?

- A. They are highly specific in nature
- B. They are proteinaceous in nature
- C. They lose activity at very high temperatures

D. They act as biocatalysts

Answers: A, B, C, D

68. Which of the following properties of Alkali metals increase as the atomic number increases?

A. Atomic radius

B. Ionic radius

C. Density (with the exception of Potassium)

D. First Ionization Enthalpy

Answers: A, B, C

69. Which of the following statements about the "Inert Pair Effect" are true?

A. It is most prominent in heavier p-block elements.

B. It involves the reluctance of s-electrons to participate in bonding.

C. It leads to the stability of lower oxidation states in Group 14 and 15.

D. It is caused by poor shielding of d and f orbitals.

Answers: A, B, C, D

70. Which of the following statements regarding KMnO_4 are correct?

A. It acts as an oxidizing agent in acidic, neutral, and alkaline media.

B. In acidic medium, it is reduced to Mn^{2+}

C. It shows colour due to d-d transitions

D. It is used as a volumetric reagent in redox titrations

Answers: A, B, D

71. Which of the following statements are correct for the Group 15 elements (Nitrogen family)?

A. The stability of +3 oxidation state increases down the group.

B. The stability of +5 oxidation state decreases down the group.

C. Nitrogen can form pπ-pπ multiple bonds.

D. Bismuth is a strong oxidizing agent in +5 oxidation state.

Answers: A, B, C, D

72. Which compound shows geometrical isomerism?

A. $\text{CH}_2=\text{CH}_2$

B. CH_3-CH_3

C. $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_3$

D. $\text{CH}_3-\text{C}\equiv\text{C}-\text{CH}_3$

Ans: C

73. The IUPAC name of $\text{CH}_3-\text{CH}(\text{OH})-\text{CH}_3$ is

A. Propan-1-ol

B. Propan-2-ol

C. Isopropyl alcohol

D. Ethanol

Ans: B

74. Which reaction forms an ether from an alkyl halide?

- A) Wurtz reaction
- B) Cannizzaro reaction
- C) Williamson synthesis
- D) Kolbe reaction

Ans: C

75. The configuration of a chiral carbon is determined using:

- A) Huckel rule
- B) CIP priority rules
- C) VSEPR theory
- D) MOT

Ans: B